Indian Railway Container Movement Analysis

Power BI Take-Home Assignment Report

1. Project Overview

This project analyzes Indian Railway container movement data using Python for data preparation and Power BI for interactive visualization. The dashboard provides journey-level and leg-level analysis to understand container movement, route performance and operational trends.

2. Data Quality & Preparation

Removed duplicate records and validated journey level data.

Converted event_time to datetime format.

Sorted movement data by journey_key and station_sequence.

Normalized destination railway divisions using station master.

Created journey level fact table (fact_od), OD summary and leg table.

Calculated journey hours, intermediate stations and route speed.

3. OD Pair Construction Methodology

The movement dataset contains one row for every station event. To perform Origin–Destination analysis, the event data was transformed into a journey-level dataset using the following steps:

1. Sorted movement data using journey_key and station_sequence.
2. Each journey_key represents one complete container journey.
3. The first event was considered as the Origin Terminal.
4. The last event was considered as the Destination Station.
5. Destination station was mapped with the Station Master to obtain the Normalized Railway Division.
6. Journey Time = Destination Event Time − Origin Event Time.
7. Intermediate Stations = Last Station Sequence − 1.
8. One record per journey was stored in fact_od and then aggregated to create the OD Summary table.

4. Dashboard Summary

Page 1 – Container Movement Overview
• KPI Cards
• Daily Trend
• Top Destination
• Origin × Destination Matrix

Page 2 – Route & Leg Analysis
• Azure Map
• Speed Category Analysis
• Leg-wise Movement Details

5. KPI Summary

• KPI	• Value
• Total Containers	• 27039
• Total OD Pairs	• 27047
• Average Journey Time	• 74.19
• Average Intermediate Stations	• 28.47
• Top Destination	• DAHANU ROAD Mumbai Central Railway Division

6. Business Insights

- Highest movement was observed from the busiest origin terminal.
- Top destination handled the maximum number of containers.
- Average journey time indicates overall network performance.
- Most journeys passed through multiple intermediate stations before reaching destination.
- Speed analysis helps identify slow route segments that may require operational attention.
- Journey-level and leg-level analysis together provide better visibility into railway container movement.

7. Business Questions Answered

• Which lanes are slowest?
Answer: Identified using average journey time and leg speed analysis.

• Which terminal generates the most long-distance NCR traffic?
Answer: Determined by filtering NCR destination divisions and comparing origin terminals.

8. Assumptions & Limitations

Haversine formula is used for approximate distance calculation.

Container identifiers were analyzed carefully because journey and transaction identifiers are not always one-to-one.

Analysis depends on the completeness of source data.

Straight-line distance may differ from actual railway track distance.

9. Conclusion

The developed Power BI solution provides an interactive view of container movement across the railway network. The dashboard supports OD analysis, KPI monitoring, route visualization and operational decision making.